1 Directed Graph

So far, we have been working with graphs with undirected edges. A directed edge is one where the end points are distinguished - one is the tail, one is the head. In particular, a directed edge is specified as an ordered pair of vertices (u, v) & is denoted by (u, v) or $u \to v$.

Definition 1.1. A directed graph $G = (V, E)$ consists of a non-empty set of nodes V & a set of directed edges E. Each edge e of E is specified by an ordered pair of vertices $u, v \in V$. A directed graph is simple if it has no loops and no multiple edges.

Example 1.2. $V = \{1, 2, 3\}$, $E = \{(1, 2), (2, 3), (3, 1)\}$

Remark 1.3. Between any two distinct vertices u, v, there can be at most one directed edge from u to v. (But both (u, v) & (v, u) are allowed, since they are in opposite direction).

In a directed graph (digraph), we distinguish between two types of degrees because edges have a direction:

- **Indegree:** The in-degree of a vertex v, denoted by $deg^-(v)$ is the number of edges entering v.
$$deg^-(v) = |\{(u, v) \in E : u \in V\}|$$
- **Outdegree:** The out-degree of a vertex v, denoted by $deg^+(v)$ is the number of edges leaving v.
$$deg^+(v) = |\{(v, u) \in E : u \in V\}|$$
Total degree of v is: $deg(v) = deg^+(v) + deg^-(v)$.

If a node has outdegree 0, it is called sink. If it has indegree 0, it is called source node (a) is Source.

Definition 1.4. A directed walk (or more simply, a walk) in a directed graph is a sequence of vertices $v_0, ..., v_k$ & edges such that $v_{i-1} \to v_i$ is an edge of G.
A directed path in a directed graph is a walk where the nodes in the walk are all different.
A directed cycle in a directed graph is a closed walk where all the vertices are different.

In the graph Fig-I, a, b, c, b, d is a walk. b, a, d is not a walk, since $b \to a$ is not an edge. $a \to b \to d$ is a path. $b \to d \to c \to b$ is a cycle.

A path or cycle in a directed graph is said to be Hamiltonian if it visits every node in the graph. $(a \to b \to d \to c)$ is the only Hamiltonian path in the graph (Fig-I). It has no Hamiltonian cycle.

1.1 Strong Connectivity

A directed graph $G = (V, E)$ is said to be strongly connected if for every pair of nodes u, v, there is a directed path from u to v.

(The Fig-I, is not strongly connected since there is no directed path from node c to node a.)

A directed graph is said to be weakly connected if the corresponding undirected graph is connected. (Fig-I is weakly connected.)

Definition 1.5. A directed graph is called a Directed Acyclic Graph (or DAG) if it does not contain any directed cycles.

Remark 1.6. A directed, at a first glance don't appear to be particularly interesting. DAGS arise in many scheduling and optimization problems.

1.2 Tournament Graphs

Suppose that n players compete in a round-robin tournament and that for every pair of players u, v, either u beats v or v beats u.

Interpreting the results of a round-robin tournament can be problematic - there might be all sorts of cycles where x beats y, y beats z, yet z beats x. Who is the best player?

Graph theory does not solve this problem but it can provide some interesting perspective.

5-node tournament Graph.

The results of a round-robin tournament can be represented with a tournament graph. This is a directed graph in which the vertices represent players and the edges indicate the outcomes of the game. In particular, an edge from u to v indicates that player u defeated player v.

Theorem 1.7. *Every tournament graph contains a directed Hamiltonian path.*

Theorem 1.8 (King Chicken Theorem)**.** *Suppose that there are n-chickens in a farm land. Chicken are rather aggressive birds that tends to establish dominance by pecking. In particular, for each pair of distinct chickens, either the first pecks the second or the second pecks the first, but not both.*

We say that Chicken u virtually pecks chicken v if either,

- *Chicken u directly pecks chicken v, or*
- *Chicken u pecks some other chicken w who in turn pecks chicken v.*

A chicken that virtually pecks every other chicken is called a King-Chicken.

(The vertices are chicken, and an edge $u \to v$ indicates that chicken u pecks chicken v.)

***Theorem**: The chicken with the largest out-degree in an n-chicken tournament is a king.*

2 Matchings

Let $G = (V, E)$ be a graph.

Definition 2.1. A subset M of edges ($M \subseteq E$) is said to be Matching/Independent if no two edges in M are incident on any vertex, or equivalently, every vertex is contained in at most one edge. (no two edges in M share a common vertex).

Types of Matching

A matching M is **perfect** if every vertex of G is incident with exactly one edge of M.

Maximum Matching: A matching that contains the largest possible number of edges in the graph.

Example 2.2. $V = \{1, 2, 3, 4\}$, $E = \{\{1, 2\}, \{2, 3\}, \{3, 4\}\}$ $M = \{\{1, 2\}, \{3, 4\}\}$ (Size is 2). It is also a maximum matching. $M_1 = \{\{1, 2\}\}$, $M_2 = \{\{2, 3\}\}$

Bipartite graph: Let G be bipartite Graph with $V_1 = \{a_1, a_2, a_3\}$, $V_2 = \{b_1, b_2, b_3\}$. $E = \{\{a_1, b_1\}, \{a_1, b_2\}, \{a_2, b_2\},$

Possible Matchings $M_1 = \{\{a_1, b_1\}, \{a_2, b_2\}, \{a_3, b_3\}\}$ $M_2 = \{\{a_1, b_2\}, \{a_3, b_3\}\}$

Alternatively, one can consider a matching of a graph M as a subgraph of G such that $d_M(v) = 1$ for all $v \in V(M)$.

A vertex 'v' is said to be saturated if $v \in V(M)$ and else unsaturated.

For a subset $S \subseteq V$, $N(S) = \bigcup_{v \in S} N(v)$ (N-denotes neighborhood of S).

Theorem 2.3 (Hall's marriage theorem, Hall, 1935)**.** *Let G be a bipartite graph with the two vertex sets V_1, V_2. Then there exists a complete matching on V_1 iff $|N(S)| \geq |S|$ for all $S \subseteq V_1$. where $N(S) = \{b \in V_2 : \exists a \in S \text{ with } (a, b) \in E\}$*

Example 2.4. $V_1 = \{a_1, a_2, a_3\}$, $V_2 = \{b_1, b_2, b_3\}$, $E = \{(a_1, b_1), (a_1, b_2), (a_2, b_2), (a_3, b_3)\}$

Now, let us check Hall's statement

- $S = \{a_1\}$, $N(S) = \{b_1, b_2\}$
- $S = \{a_2\}$, $N(S) = \{b_2\}$
- $S = \{a_3\}$, $N(S) = \{b_3\}$
- $S = \{a_1, a_2\}$, $N(S) = \{b_1, b_2\}$
- $S = \{a_1, a_3\}$, $N(S) = \{b_1, b_2, b_3\}$
- $S = \{a_1, a_2, a_3\}$, $N(S) = \{b_1, b_2, b_3\}$.

Matching is $M = \{(a_1, b_1), (a_2, b_2), (a_3, b_3)\}$.

Proof of Hall's Theorem. Let $|V_1| = k$, and our proof will be by induction on k. If $k = 1$, the proof is trivial. Let $G = V_1 \cup V_2$ be such that the result holds for any graph with strictly smaller V_1.

Case 1: Suppose that $|N(S)| \geq |S| + 1$ for all $S \subsetneq V_1$. Then choose $(v, w) \in E \cap (V_1 \times V_2)$ & consider the induced subgraph $G' = \langle V - \{v, w\} \rangle$. Since we have removed only w from V_2 and that $|N(S)| \geq |S| + 1$ for $S \subsetneq V_1$, we get $|N(S')| \geq |S'|$ in G'. Thus there is a complete matching M in G' by induction hypothesis & $M \cup \{(v, w)\}$ is a complete matching on V_1 in G as desired.

Case 2: If the above is not true, there exists $A \subsetneq V_1$ such that $N(A) = B$ & $|A| = |B|$. Then by induction hypothesis, there is a complete matching M_A on A in the induced subgraph $\langle A \cup B \rangle$. (Trivially Hall's condition holds i.e., for all $S \subset A$, $|N(S) \cap B| = |N(S)| \geq |S|$).

Let $G' = G - \langle A \cup B \rangle$. Let $S \subset V_1 \setminus A$. Suppose if $|N'(S)| < |S|$, where $N'(S) = N(S) \cap (V_2 \setminus B)$. Then we have that $N(S \cup A) = [N(S) \cap (V_2 \setminus B)] \cup B$. and hence $|N(S \cup A)| = |N(S) \cap (V_2 \setminus B)| + |B| < |S| + |A|$. But $|N(S \cup A)| \geq |S \cup A| = |S| + |A|$. This is a contradiction.

Hence G' also satisfies Hall's condition and by induction hypothesis G' has a complete matching M' on $V_1 \setminus A$. Thus we have a complete matching $M := M_A \cup M'$ on V_1 in G. □

Proposition 2.5. *Let $d \geq 1$. Let G be a bipartite graph on $V_1 \sqcup V_2$ such that $|N(S)| \geq |S| - d$ for all $S \subseteq V_1$. Then G has a matching with at least $|V_1| - d$ edges.*

Proof. Define $G' = V_1 \cup V_2'$ where $V_2' := V_2 \cup \{1, 2, ..., d\}$ and edge set as $E(G) \cup (V_1 \times [d])$. Then it is easy to see that Hall's ($|N'(S)| \geq |S|$) holds and hence there is a complete matching M of V_1 in G'. Now, if we remove the edges in M incident on $\{1, 2, ..., d\}$, we get a matching with at least $|V_1| - d$ edges as required. $\square$

Definition 2.6 (Independent sets and covers)**.** An **independent set** of vertices is $S \subseteq V$ such that no two vertices in S are adjacent.

A subset of vertices $S \subseteq V$ is a **vertex cover** if every edge in G is incident to at least one vertex in S.

An **edge cover** is a set of edges $E' \subseteq E$ such that every vertex is contained in at least one $e \in E'$.

Definition 2.7 (Independence number and cover number)**.**

$$\begin{aligned}
\alpha(G) &= \max\{|S| : S \text{ independent vertex set}\} \\
\alpha'(G) &= \max\{|M| : M \text{ independent edge set (Matching)}\} \\
\beta(G) &= \min\{|S| : S \text{ vertex cover}\} \\
\beta'(G) &= \min\{|E'| : E' \text{ edge cover}\}
\end{aligned}$$

We first derive some trivial relations between the four quantities. If M is a maximal matching, then to cover each edge we need distinct vertices and hence the vertex cover should have size at least $|M|$. (i.e., $\alpha'(G) \leq \beta(G)$).

Also: $\alpha(G) \leq \beta'(G)$ (to cover vertices of an independent set, we need distinct edges).

Lemma 2.8. *Let G be a graph. $S \subseteq V$ is an independent set iff S^c is a vertex cover.*

Proof. Assume S is independent. Take any edge $(u, v) \in E$. Since u, v are adjacent, they cannot both lie in the independent set S. Thus at least one of u or v is not in S, i.e., at least one of u, v lies in S^c. Because this holds for every edge, every edge has an end point in S^c. Hence S^c is a vertex cover.

Conversely. Assume S^c is a vertex cover. Assume S is not independent. Then there exist $u, v \in S$ with $(u, v) \in E$. But as S^c is a vertex cover, every edge has an end point in S^c. $\Rightarrow (u, v)$ has end points outside S, not possible. $\square$

Corollary 2.9. $\alpha(G) + \beta(G) = n = |V|$.

Example 2.10. Let $G = (V, E)$, where $V = \{1, 2, 3\}, E = \{(1, 2), (2, 3)\}$.

1. Independent set of vertices: $\{1, 3\}$. ($\alpha(G) = 2$)
2. Vertex Cover: $\{2\} \leftarrow$ min vertex cover. ($\beta(G) = 1$)
3. Edge cover: $\{(1, 2), (2, 3)\}$. ($\beta'(G) = 2$)

Observe: $\alpha(G) + \beta(G) = 2 + 1 = 3 = n$.

Example 2.11. $V = \{A, B, C, D\}$, $E = \{AB, AC, BD\}$.

- Independent set: $\{CD\}, \{AD\}, \{BC\}$. ($\alpha(G) = 2$)
- Vertex Cover: $\{A, B\}, \{A, D\}$. ($\beta(G) = 2$)

- Edge Cover: e.g., $\{AC, BD\}$. ($\beta'(G) = 2$)

Theorem 2.12 (König, Egerváry, 1931)**.** *For a Bipartite graph G, $\alpha'(G) = \beta(G)$.*

Proof. We will show that for a minimal vertex cover Q, there exists a matching of size at least $|Q|$. Partition Q into $A := Q \cap V_1$ and $B := Q \cap V_2$. Let H_1 be the induced subgraph on $\langle A \cup (V_2 \setminus B)\rangle$ and H_2 be the induced subgraph on $\langle (V_1 \setminus A) \cup B\rangle$. If we show that there is a complete matching on A in H_1 and a complete matching on B in H_2, we have a matching of size at least $|A| + |B| = |Q|$.

Also, note that it suffices to show that there is a complete matching on A in H_1 because we can reverse the role of A & B, and apply the same argument to H_2.

Since $A \cup B$ is a vertex cover, there can not be an edge between $V_1 \setminus A$ & $V_2 \setminus B$. Suppose for some $S \subseteq A$ we have that $|N_{H_1}(S)| < |S|$ (where $N_{H_1}(S) \subseteq (V_2 - B)$). $Q' = (Q \setminus S) \cup N_{H_1}(S)$ is also a vertex cover. By choice of Q, $(|N_{H_1}(S)| < |S|) \Rightarrow Q'$ is a smaller vertex cover than Q, contradicting the minimality of Q. Hence $\Rightarrow |N_{H_1}(S)| \geq |S|$ for all $S \subseteq A$. $\Rightarrow$ Hall's criterion satisfies $\Rightarrow$ there is a complete matching for A in H_1. This matching is of size at least $|A|$. (A symmetric argument shows a matching of size at least $|B|$ in H_2, and their union is the desired matching of size $|Q|$). ☐

Theorem 2.13 (Gallai, 1959)**.** *If G is a graph without isolated vertices, then $\alpha'(G) + \beta'(G) = n = |V|$.*

Proof. Suppose M is a maximal matching. Then $S = V \setminus V(M)$ is also an independent set. Indeed, if there are edges between vertices of S, then such edges can be added to M and one can obtain a larger matching. Hence S is independent.

Construct an edge cover Q as follows: Add all the edges in M to Q and for each $v \in S$, add one of its adjacent edges to Q. Thus Q is an edge cover, so $\beta'(G) \leq |Q| = |M| + |S|$. Since $V(M) \sqcup S = V$, we have $n = |V(M)| + |S| = 2|M| + |S|$. We have $\alpha'(G) \geq |M|$. $\alpha'(G) + \beta'(G) \leq \alpha'(G) + |M| + |S|$. (This part of the proof in the notes seems complex) Let's use the provided text: $|Q| = |M| + |S|$. $\alpha'(G) + \beta'(G) \leq |M| + |Q| = |M| + (|M| + |S|) = 2|M| + |S| = n$. $\Rightarrow \alpha'(G) + \beta'(G) \leq n$.

Now, let Q be a minimal edge cover. Then Q cannot contain a path of length more than 2. Else, by removing the middle edge in a path of length at least 3, we can obtain a smaller edge cover. (Using a result, if G does not contain a path of length more than 2, then its connected components are all star graphs $K_{1,k}$). $\Rightarrow Q$ is a graph consisting of star components. If $C_1, ..., C_k$ are the components of Q. Then $V(C_1) \cup ... \cup V(C_k) = V$ and $E(C_1) \cup ... \cup E(C_k) = Q$. Now choose a matching $M = \{e_1, e_2, ..., e_k\}$ by selecting one edge from each component. (Since C_i's are disjoint), M is a matching. $\alpha'(G) + \beta'(G) \geq |M| + |Q| = k + \sum_{i=1}^{k} |E(C_i)|$. Since C_i is a star, $|V(C_i)| = |E(C_i)| + 1$. Summing over all components: $\sum |V(C_i)| = \sum(|E(C_i)| + 1) = (\sum |E(C_i)|) + k = |Q| + |M|$. Since $\sum |V(C_i)| = n$, we have $n = |Q| + |M|$. Thus, $\alpha'(G) + \beta'(G) \geq |M| + |Q| = n$. ☐

2.1 Algorithm: Gale-Shapley Algorithm

In 1962, Gale-Shapley proposed an algorithm to achieve stable matching and this probably the best known of such algorithm. Along with Roth, Lloyd Shapley was awarded Noble prize in economics in 2012. (stable allocation pb)

Background: The *Gale–Shapley Algorithm*, also known as the *Deferred Acceptance Algorithm*, was proposed in 1962 by David Gale and Lloyd Shapley. It is designed to find a **stable matching**

between two equally sized sets of agents typically referred to as "men" and "women" based on their preferences for one another.

In recognition of this foundational work, Lloyd Shapley (together with Alvin E. Roth) was awarded the *2012 Nobel Prize in Economic Sciences* for the theory and practice of stable allocations and market design.

The Stable Marriage Problem: Given two disjoint sets:

$$M = \{m_1, m_2, \ldots, m_n\} \quad \text{and} \quad W = \{w_1, w_2, \ldots, w_n\}$$

Each man m_i has a **preference list** ranking all women from most to least preferred. Similarly, each woman w_j ranks all men from most to least preferred.

A **matching** is a one-to-one correspondence between M and W. A matching is said to be *stable* if there is no *blocking pair* (m, w) such that: - m prefers w over his current partner, and - w prefers m over her current partner.

If such a pair exists, the matching is *unstable*.

Goal: To find a **stable matching** between M and W.

Algorithm (Gale–Shapley Deferred Acceptance): The algorithm proceeds in rounds, where one side (say, the men) "propose" and the other side (the women) "accept or reject" tentatively.

1. Initially, all men and women are free (unmatched).
2. While there exists a free man m who has not yet proposed to every woman on his preference list:
   (a) m proposes to the most preferred woman w on his list who has not yet rejected him.
   (b) If w is free, she **tentatively accepts** the proposal from m.
   (c) If w is already matched to some man m':
      - If w prefers m' over m, she **rejects** m.
      - If w prefers m over m', she **rejects** m' and tentatively accepts m.
3. Repeat until every man is matched (i.e., no free men remain).

The algorithm terminates with a **stable matching**.

Example: Suppose

$$M = \{m_1, m_2\}, \quad W = \{w_1, w_2\}$$

Preferences:

Men's Preferences	Women's Preferences
$m_1 : w_1 > w_2$	$w_1 : m_2 > m_1$
$m_2 : w_1 > w_2$	$w_2 : m_1 > m_2$

Step 1: Both m_1 and m_2 propose to w_1. w_1 prefers m_2, so she accepts m_2 and rejects m_1.

Step 2: m_1 now proposes to w_2. w_2 is free, so she accepts m_1.

Final Matching:

$$M = \{(m_1, w_2), (m_2, w_1)\}$$

This matching is **stable**.

Properties:

- The algorithm always terminates after a finite number of steps.
- The resulting matching is **stable**.
- The matching is **optimal for the proposing side** (e.g., men) and **pessimal for the receiving side** (e.g., women).
- The outcome may differ if the women propose instead; that would yield a **woman-optimal** stable matching.

Time Complexity: Each man proposes to each woman at most once. Hence, the worst-case running time is:

$$O(n^2)$$

Applications: Beyond the classical marriage problem, the Gale–Shapley algorithm has many real-world applications, such as:

- College admissions
- Hospital–residents assignments
- School choice systems
- Organ donor matching

Reference: Section 3, for further discussion of matchings and stability. Please See D. B. West, *Introduction to Graph Theory*.

3 Graph Coloring

Definition 3.1 (Coloring of a graph)**.** Let $G = (V, E)$ be a graph. A graph is k-colorable if $\exists c : V \to \{1, 2, ..., k\}$ such that $c(u) \neq c(v)$ if $(u, v) \in E$. The chromatic number $\chi(G)$ is defined as

$$\chi(G) := \inf\{k : G \text{ is } k\text{-colorable}\}$$

Example 3.2.

1. **Empty Graph** No edges. All vertices can have the same color. $\chi(G) = 1$.
2. **Complete Graph** K_n Every pair of vertices is adjacent. $\chi(G) = n$.
3. **Cyclic Graph** C_n If n is even, $\chi(C_n) = 2$. If n is odd, $\chi(C_n) = 3$.
4. **Bipartite Graph** Vertices $V = V_1 \cup V_2$. $\chi(G) = 2$ (unless it has no edges, in which case $\chi(G) = 1$).

3.1 Chromatic Polynomials

Let G be a graph with n-vertices. Let $P(G, q)$, $q \in \mathbb{N}$ be the number of ways of coloring (properly) a graph with q colours.

$$P(G, q) := |\{c : V \to \{1, 2, ..., q\} : c(u) \neq c(v) \forall (u, v) \in E\}|$$

If G is a graph & $e = (u, v)$ is an edge of G, then

- $G - e$ denotes the graph with the edge 'e' removed.
- G/e denotes the graph with e contracted. (merge the two end points u & v into a single vertex, remove any loops that may be created.)But will not matter to us as proper colorings of a multi-graph and its corresponding simple graph are the same.

Proposition 3.3. *For any edge $e = (u, v) \in G$,*

$$P(G, q) := P(G - e, q) - P(G/e, q), \quad q \in \mathbb{N}$$

Proof. Observe that q-coloring of G is a proper q-coloring of $G - e$ in which u, v receive distinct colours. and any coloring of $G - e$ in which u, v receive same colours is a proper q-coloring of G/e. □

Remark 3.4. If $E = \emptyset$, then $P(G, q) = q^n$. We shall show that $P(G, q)$ is a polynomial. Hence we shall define $P(G, x)$, $x \in \mathbb{R}$ to be the polynomial such that $P(G, q)$ is the number of proper q-colorings of G, $q \in \mathbb{N}$.

Lemma 3.5. *1. $P(G, x)$ is a monic polynomial of degree n.*

2. $\chi(G) = \min\{k \in \mathbb{N} : P(G, k) > 0\}$.

3. Now, let us assume that $P(G, x) = \sum_{i=0}^{n} a_i x^i$. Then we have that $a_{n-1} = -|E|$.

Proof of 1 and 3. Let m be the number of edges in G. **Base Case:** $m = 0$; G has no edge $\Rightarrow P(G, x) = x^n$. This is a polynomial of degree n, and leading coefficient 1. $a_{n-1} = 0 = -|E|$.

Induction hypothesis: Fix n (= number of vertices). Assume the claim holds for all graphs on n vertices having fewer than m edges.

Inductive Step: Let G be a graph with m edges. Pick any edge $e = (u, v)$ of G. Then

$$P(G, x) = P(G - e, x) - P(G/e, x)$$

$G - e$ has the same vertex set as G, hence has n-vertices with '$m-1$' edges. By induction hypothesis $P(G - e, x)$ is a polynomial of degree n & monic. $P(G - e, x) = x^n - (m - 1)x^{n-1} + \dots$

G/e is the graph with at most $(n - 1)$ vertices.

- a) If the contraction produces a loop at the merged vertex, then G/e has no proper colorings for any x & $P(G/e, x) = 0$. So it does not effect the leading term of $P(G, x)$.
- b) If no loop is created, then G/e is a simple graph on exactly $(n - 1)$ vertices. By applying the induction hypothesis (on the number of vertices) we know that $P(G/e, x)$ is a poly. of degree $n - 1$ with leading co-efficient 1.

So, $P(G/e, x) = x^{n-1} + \dots$

$$P(G, x) = (x^n - (m-1)x^{n-1} + ...) - (x^{n-1} + ...)$$

$$P(G, x) = x^n - (m - 1 + 1)x^{n-1} + ... = x^n - mx^{n-1} + ...$$

Thus $P(G, x)$ is monic of degree n, and $a_{n-1} = -m = -|E|$. $\square$

Lemma 3.6. *If G has k-components, $G_1, ..., G_k$ then*

$$P(G, x) = \prod_{i=1}^{k} P(G_i, x)$$

and further

$$a_o, a_1 = ... = a_{k\text{-}1} = 0, |a_k| > 0$$

Lemma 3.7. *A graph G with n-vertices is a tree iff $P(G, x) = x(x-1)^{n-1}$.*

Proof. ($\Rightarrow$) Let T be a tree on 'n' vertices. Pick any vertex as root. Let us colour it first; there are x choices. Then colour its neighbour: each neighbour cannot take the root's colour. Continue recursively down the tree. Each new vertex has $(x-1)$ choices because it is adjacent to exactly one already-coloured vertex. $\Rightarrow P(T, x) = x(x-1)^{n-1}$.

($\Leftarrow$) Conversely, suppose G has n-vertices and $P(G, x) = x(x-1)^{n-1}$. $\Rightarrow$ $deg(P(G, x)) = n$. Observe that for a Chromatic poly. $P(G, x) = x^n - mx^{n-1} + ...$ where m = number of edges of G. From the expansion:

$$x(x-1)^{n-1} = x(x^{n-1} - (n-1)x^{n-2} + ...) = x^n - (n-1)x^{n-1} + ...$$

Comparing the coefficients of x^{n-1}, we get $m = (n-1)$. $\Rightarrow G$ has exactly $(n-1)$ edges. A graph G with n vertices and $n-1$ edges is a tree iff it is connected. If G is not connected, $P(G, 1)$ would be 0, but $P(G, x)$ has components G_i, so $P(G, x) = \prod P(G_i, x)$. If G is not connected, $P(G, 1) = 0$. $P(T, 1) = 1(1-1)^{n-1} = 0$ (for $n > 1$). Also $P(G, 2) = 2(2-1)^{n-1} = 2 > 0$ (if $n \geq 1$), so G is 2-colorable, i.e., bipartite. A graph with n vertices, $n-1$ edges, and no cycles (which it must not have, otherwise $P(C_k, x)$ would be a factor) must be a tree. $\square$